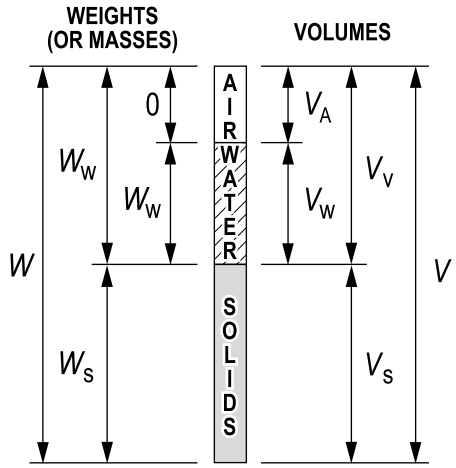
Civil Engineering

Geotechnical

Phase Relationships



PHASE DIAGRAM

Volume of voids

$$V_V = V_A + V_W$$

Total unit weight

$$\gamma = \frac{W}{V}$$

Saturated unit weight

$$\gamma_{sat} = \frac{(G_s + e)\gamma_w}{1 + e}$$

$$\gamma_W = 62.4 \text{ lb/ft}^3 \text{ or } 9.81 \text{ kN/m}^3$$

Effective (submerged) unit weight

$$\gamma' = \gamma_{sat} - \gamma_W$$

Unit weight of solids

$$\gamma_S = \frac{W_S}{V_S}$$

Dry unit weight

$$\gamma_D = \frac{W_S}{V}$$

Water content (%)

$$\omega = \frac{W_W}{W_S} \times 100$$

Specific gravity of soil solids

$$G_S = (W_S/V_S)/\gamma_W$$

Void ratio

$$e = \frac{V_V}{V_S}$$

Porosity

$$n = \frac{V_V}{V} = \frac{e}{1 + e}$$

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Degree of saturation (%)

$$S = \frac{V_W}{V_V} \times 100$$

$$S = \frac{\omega G_S}{e}$$

Relative density

$$D_r = [(e_{max} - e)/(e_{max} - e_{min})] \times 100$$
$$= [(\gamma_{D\ field} - \gamma_{D\ min})/(\gamma_{D\ max} - \gamma_{D\ min})][\gamma_{D\ max}/\gamma_{D\ field}] \times 100$$

Relative compaction (%)

$$RC = (\gamma_{D\ field}/\gamma_{D\ max}) \times 100$$

Plasticity index

$PI = LL - PL$

LL = liquid limit

PL = plastic limit

Coefficient of uniformity

$$C_U = D_{60}/D_{10}$$

Coefficient of concavity (or curvature)

$$C_C = (D_{30})^2/(D_{10} \times D_{60})$$

where

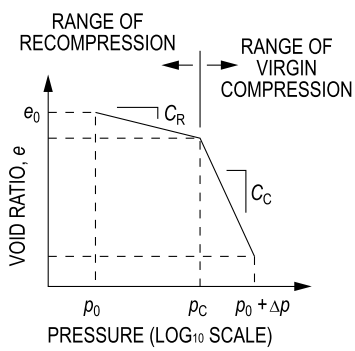
$D_{\#\#}$ = grain diameter (mm) corresponding to ##% passing by weight

Factor of safety against seepage liquefaction

$FS_s = i_c/i_e$

$i_c = (\gamma_{sat} - \gamma_W)/\gamma_W$

i_e = seepage exit gradient



SOIL CONSOLIDATION CURVE OVER CONSOLIDATED CLAY

where

e_0 = initial void ratio (prior to consolidation)

Δe = change in void ratio

p_0 = initial effective consolidation stress σ'_0

p_c = past maximum consolidation stress σ'_c

Δp = induced change in consolidation stress at center of consolidating stratum

$\Delta p = I\, q_s$

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where

I = Stress influence value at center of consolidating stratum

q_s = applied surface stress causing consolidation

If $(p_o < p_c$ and $p_o + \Delta p < p_c)$, then $\Delta H = \frac{H_o}{1+e_o}\left[C_R \log\frac{p_o+\Delta p}{p_o}\right]$

If $(p_o \geq p_c$ and $p_o + \Delta p \geq p_c)$, then $\Delta H = \frac{H_o}{1+e_o}\left[C_C \log\frac{p_o+\Delta p}{p_o}\right]$

If $p_o < p_c < (p_o + \Delta p)$, then $\Delta H = \frac{H_o}{1+e_o}\left[C_R \log\frac{p_c}{p_o} + C_C \log\frac{p_o+\Delta p}{p_c}\right]$

ΔH = change in thickness of soil layer

Compression index
In virgin compression range: $C_C = \Delta e/\Delta \log p$
By correlation to liquid limit: $C_C = 0.009\ (LL - 10)$

Recompression index
In recompression range: $C_R = \Delta e/\Delta \log p$
By correlation to compression index, C_C: $C_R = C_C/6$

Ultimate consolidation settlement in soil layer

$S_{ULT} = \varepsilon_v H_S$

where H_S = thickness of soil layer

$\varepsilon_v = \Delta e_{TOT}/(1 + e_0)$

where Δe_{TOT} = total change in void ratio due to recompression and virgin compression

Approximate settlement (at time $t = t_C$)

$S_T = U_{AV} S_{ULT}$

where

U_{AV} = average degree of consolidation

t_C = elapsed time since application of consolidation load

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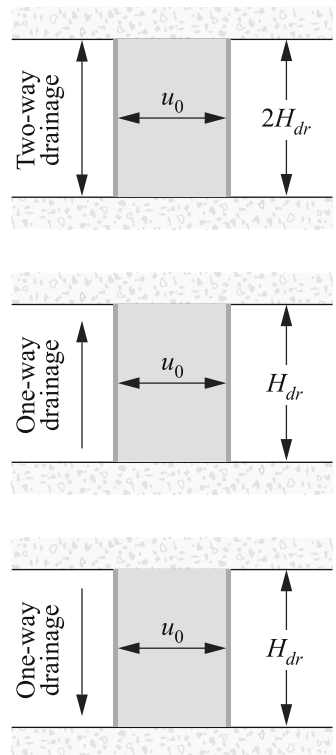
Variation of time factor with degree of consolidation*

U (%)	T_v	U (%)	T_v	U (%)	T_v
0	0	34	0.0907	68	0.377
1	0.00008	35	0.0962	69	0.390
2	0.0003	36	0.102	70	0.403
3	0.00071	37	0.107	71	0.417
4	0.00126	38	0.113	72	0.431
5	0.00196	39	0.119	73	0.446
6	0.00283	40	0.126	74	0.461
7	0.00385	41	0.132	75	0.477
8	0.00502	42	0.138	76	0.493
9	0.00636	43	0.145	77	0.511
10	0.00785	44	0.152	78	0.529
11	0.0095	45	0.159	79	0.547
12	0.0113	46	0.166	80	0.567
13	0.0133	47	0.173	81	0.588
14	0.0154	48	0.181	82	0.610
15	0.0177	49	0.188	83	0.633
16	0.0201	50	0.197	84	0.658
17	0.0227	51	0.204	85	0.684
18	0.0254	52	0.212	86	0.712
19	0.0283	53	0.221	87	0.742
20	0.0314	54	0.230	88	0.774
21	0.0346	55	0.239	89	0.809
22	0.0380	56	0.248	90	0.848
23	0.0415	57	0.257	91	0.891
24	0.0452	58	0.267	92	0.938
25	0.0491	59	0.276	93	0.993
26	0.0531	60	0.286	94	1.055
27	0.0572	61	0.297	95	1.129
28	0.0615	62	0.307	96	1.219
29	0.0660	63	0.318	97	1.336
30	0.0707	64	0.329	98	1.500
31	0.0754	65	0.340	99	1.781
32	0.0803	66	0.352	100	∞
33	0.0855	67	0.364		

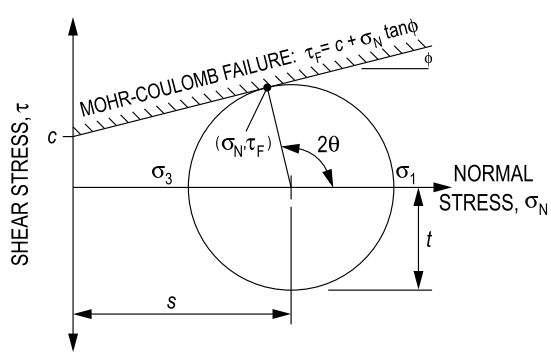
*u_0 constant with depth.



Different types of drainage with u_0 constant

where time factor is $T_v = \dfrac{c_v t}{H_{dr}^2}$

Source: Das, Braja M., *Fundamentals of Geotechnical Engineering*, Cengage Learning (formerly Brooks/Cole), 2000.



where

s = mean normal stress

t = maximum shear stress

σ_1 = major principal stress

σ_3 = minor principal stress

θ = orientation angle between plane of existing normal stress and plane of major principal stress

Total normal stress

$$\sigma_N = P/A$$

where

P = normal force

A = cross-sectional area over which force acts

Effective stress

$$\sigma' = \sigma - u$$

$$u = h_u \gamma_w$$

where h_u = uplift or pressure head

Shear stress

$$\tau = T/A$$

where T = shearing force

Shear stress at failure

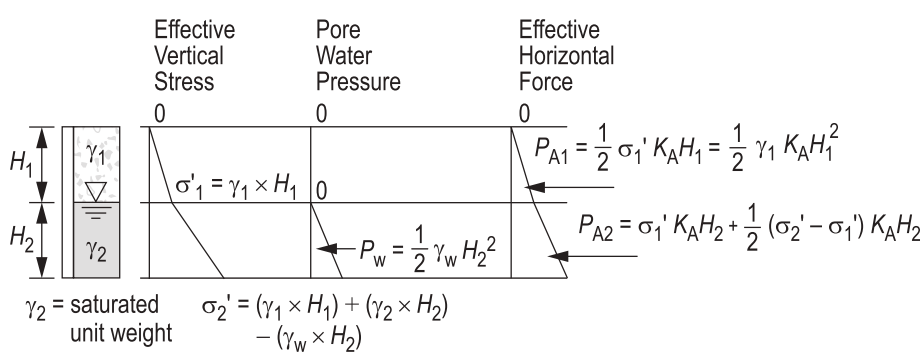
$$\tau_F = c + \sigma_N \tan \phi$$

where

c = cohesion

ϕ = angle of internal friction

Horizontal Stress Profiles and Forces



Active forces on retaining wall per unit wall length (as shown):

K_A = Rankine active earth pressure coefficient (smooth wall, $c = 0$, level backfill) = $\tan^2 (45° - \phi/2)$

Passive forces on retaining wall per unit wall length (similar to the active forces shown):

K_P = Rankine passive earth pressure coefficient (smooth wall, $c = 0$, level backfill) = $\tan^2 (45° + \phi/2)$

At-rest forces on wall per unit length of wall

K_0 = at-rest earth pressure coefficient (smooth wall, $c = 0$, level backfill)

$K_0 \approx 1 - \sin \phi$ for normally consolidated soil

$K_0 = (1 - \sin \phi)\, \mathrm{OCR}^{\sin \phi}$ for overconsolidated soil where

OCR = overconsolidation ratio

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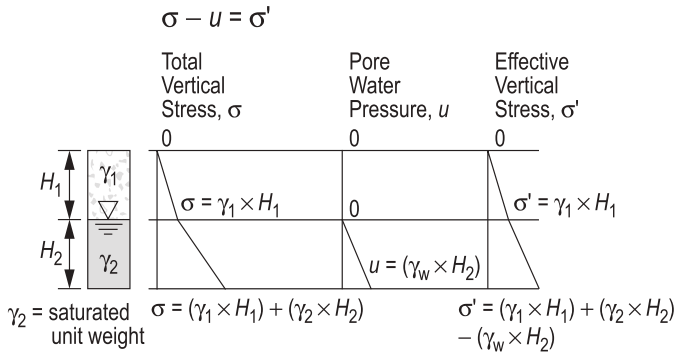
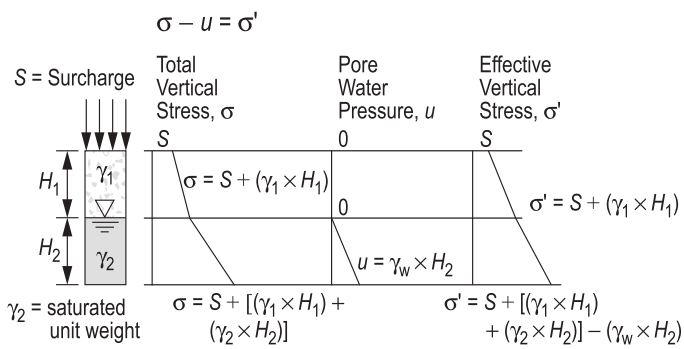
Vertical Stress Profiles



Vertical Stress Profiles with Surcharge



Ultimate Bearing Capacity

$$q_{\text{ULT}} = cN_c + \gamma' D_f N_q + \frac{1}{2}\gamma' BN_\gamma$$

where

N_c = bearing capacity factor for cohesion
N_q = bearing capacity factor for depth
N_γ = bearing capacity factor for unit weight
D_f = depth of footing below ground surface
B = width of strip footing

Retaining Walls

$$FS_{\text{overturning}} = \frac{\sum M_R}{M_O}$$

$$FS_{\text{sliding}} = \frac{\sum F_R}{\sum F_D}$$

$$FS_{\text{sliding}} = \frac{(\sum V)\tan\delta + BC_a + P_p}{P_a \cos\alpha}$$

$$FS_{\text{bearing capacity}} = \frac{q_{\text{ULT}}}{q_{\text{toe}}}$$

$$q_{\text{toe}} = \frac{\sum V}{B}\left(1 + \frac{6e}{B}\right)$$

$$e = \frac{B}{2} - \left(\frac{\sum M_R - M_O}{\sum V}\right)$$

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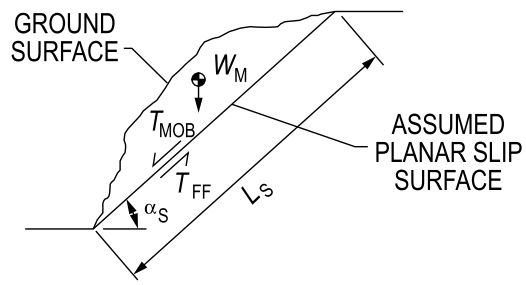
where

e = eccentricity
B = width of base
M_R = resisting moment
M_O = overturning moment
F_R = resisting forces
F_D = driving forces
V = vertical forces
$\delta = k_1\phi_2$
$C_a = k_2C_2$
k_1 and k_2 are given, ranging from 1/2 to 2/3



SLOPE FAILURE ALONG PLANAR SURFACE

where

FS = factor of safety against slope instability
= T_{FF}/T_{MOB}
T_{FF} = available shearing resistance along slip surface
= $cL_S + W_M \cos \alpha_S \tan \phi$

where
c = cohesion
ϕ = angle of internal friction

T_{MOB} = mobilized shear force along slip surface
= $W_M \sin \alpha_S$
L_S = length of assumed planar slip surface
W_M = weight of soil above slip surface
α_S = angle of assumed slip surface with respect to horizontal

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AASHTO Soil Classification

GENERAL CLASSIFICATION	GRANULAR MATERIALS (35% OR LESS PASSING 0.075 SIEVE)							SILT-CLAY MATERIALS (MORE THAN 35% PASSING 0.075 SIEVE)			
GROUP CLASSIFICATION	A-1		A-3	A-2				A-4	A-5	A-6	A-7-5 A-7-6
	A-1-a	A-1-b		A-2-4	A-2-5	A-2-6	A-2-7				
SIEVE ANALYSIS, PERCENT PASSING:											
2.00 mm (No. 10)	≤ 50	–	–	–	–	–	–	–	–	–	–
0.425 mm (No. 40)	≤ 30	≤ 50	≥ 51	–	–	–	–	–	–	–	–
0.075 mm (No. 200)	≤ 15	≤ 25	≤ 10	≤ 35	≤ 35	≤ 35	≤ 35	≥ 36	≥ 36	≥ 36	≥ 36
CHARACTERISTICS OF FRACTION PASSING 0.425 SIEVE (No. 40):											
LIQUID LIMIT	–		–	≤ 40	≥ 41	≤ 40	≥ 41	≤ 40	≥ 41	≤ 40	≥ 41
PLASTICITY INDEX *	6 max		NP	≤ 10	≤ 10	≥ 11	≥ 11	≤ 10	≤ 10	≥ 11	≥ 11
USUAL TYPES OF CONSTITUENT MATERIALS	STONE FRAGM'TS, GRAVEL, SAND		FINE SAND	SILTY OR CLAYEY GRAVEL AND SAND				SILTY SOILS		CLAYEY SOILS	
GENERAL RATING AS A SUBGRADE	EXCELLENT TO GOOD							FAIR TO POOR			

*Plasticity index of A-7-5 subgroup is equal to or less than LL – 30. Plasticity index of A-7-6 subgroup is greater than LL – 30.
NP = Non-plastic (use "0"). Symbol "–" means that the particular sieve analysis is not considered for that classification.

If the soil classification is A4-A7, then calculate the group index (GI) as shown below and report with classification. The higher the GI, the less suitable the soil. Example: A-6 with GI = 15 is less suitable than A-6 with GI = 10.

$$GI = (F - 35)[0.2 + 0.005(LL - 40)] + 0.01(F - 15)(PI - 10)$$

where: F = Percent passing No. 200 sieve, expressed as a whole number. This percentage is based only on the material passing the No. 200 sieve.
LL = Liquid limit
PI = Plasticity index

If the computed value of GI < 0, then use GI = 0.

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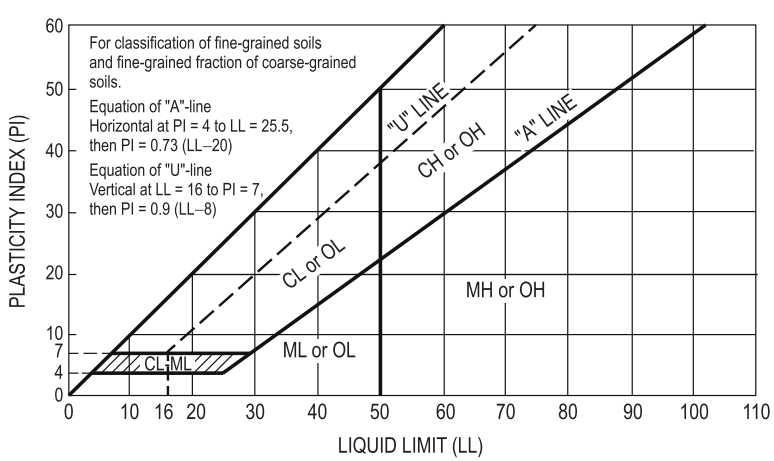
ASTM D2487-11 Standard Practice for Classification of Soils for Engineering Purposes (Unified Soil Classification System)

Criteria for Assigning Group Symbols and Group Names Using Laboratory Tests[A]				Soil Classification	
				Group Symbol	Group Name[B]
COARSE-GRAINED SOILS	Gravels (more than 50% of coarse fraction retained on No. 4 sieve)	Clean Gravels (Less than 5% fines[C])	Cu ≥ 4 and 1 ≤ Cc ≤ 3[D]	GW	Well-graded gravel[E]
			Cu < 4 and/or [Cc < 1 or Cc > 3][D]	GP	Poorly graded gravel[E]
		Gravels with Fines (More than 12% fines[C])	Fines classify as ML or MH	GM	Silty gravel[E, F, G]
			Fines classify as CL or CH	GC	Clayey gravel[E, F, G]
More than 50% retained on No. 200 sieve	Sands (50% or more of coarse fraction passes No. 4 sieve)	Clean Sands (Less than 5% fines[H])	Cu ≥ 6 and 1 ≤ Cc ≤ 3[D]	SW	Well-graded sand[I]
			Cu < 6 and/or [Cc < 1 or Cc > 3][D]	SP	Poorly graded sand[I]
		Sands with Fines (More than 12% fines[H])	Fines classify as ML or MH	SM	Silty sand[F, G, I]
			Fines classify as CL or CH	SC	Clayey sand[F, G, I]
FINE-GRAINED SOILS	Silts and Clays	inorganic	PI > 7 and plots on or above "A" line[J]	CL	Lean clay[K, L, M]
	Liquid limit less than 50		PI < 4 or plots below "A" line[J]	ML	Silt[K, L, M]
		organic	Liquid limit – oven dried/Liquid&#10 < 0.75	OL	Organic clay[K, L, M, N] / Organic silt[K, L, M, O]
50% or more passes the No. 200 sieve	Silts and Clays	inorganic	PI plots on or above "A" line	CH	Fat clay[K, L, M]
	Liquid limit 50 or more		PI plots below "A" line	MH	Elastic silt[K, L, M]
		organic	Liquid limit – oven dried/Liquid&#10 < 0.75	OH	Organic clay[K, L, M, P] / Organic silt[K, L, M, Q]
HIGHLY ORGANIC SOILS	Primarily organic matter, dark in color, and organic odor			PT	Peat

[A]Based on the material passing the 3-in. (75-mm) sieve.
[B]If field sample contained cobbles or boulders, or both, add "with cobbles or boulders, or both" to group name.
[C]Gravels with 5 to 12% fines require dual symbols:
GW-GM well-graded gravel with silt
GW-GC well-graded gravel with clay
GP-GM poorly graded gravel with silt
GP-GC poorly graded gravel with clay

[D] $Cu = D_{60}/D_{10}$ $\quad Cc = \dfrac{(D_{30})^2}{D_{10} \times D_{60}}$

[E]If soil contains ≥ 15% sand, add "with sand" to group name.
[F]If fines classify as CL-ML, use dual symbol GC-GM, or SC-SM.
[G]If fines are organic, add "with organic fines" to group name.
[H]Sands with 5 to 12% fines require dual symbols:
SW-SM well-graded sand with silt
SW-SC well-graded sand with clay
SP-SM poorly graded sand with silt
SP-SC poorly graded sand with clay
[I]If soil contains ≥ 15% gravel, add "with gravel" to group name.
[J]If Atterberg limits plot in hatched area, soil is a CL-ML, silty clay.
[K]If soil contains 15 to 30% plus No. 200, add "with sand" or "with gravel", whichever is predominant.
[L]If soil contains ≥ 30% plus No. 200, predominantly sand, add "sandy" to group name.
[M]If soil contains ≥ 30% plus No. 200, predominantly gravel, add "gravelly" to group name.
[N]PI ≥ 4 and plots on or above "A" line.
[O]PI < 4 or plots below "A" line.
[P]PI plots on or above "A" line.
[Q]PI plots below "A" line.



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